

## ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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# 1. GATE BT 2025

## A. Formulae

### 1.1.1 Exponential Series

- Taylor (Maclaurin) Series of  $e^x$ :

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (1.1.1.1)$$

The series converges absolutely for all  $x \in \mathbb{R}$  by the ratio test:

$$L = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1 \quad (1.1.1.2)$$

### 1.1.2 Intersection of Conics and Common Chord

- Line-Conic Intersection: The points of intersection of a line  $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$  ( $\kappa \in \mathbb{R}$ ) with a conic section  $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$  are given by:

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.1.2.1)$$

where the parameter values  $\kappa_i$  are:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.1.2.2)$$

with  $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$ .

- Matrix Formulation of Intersection Points: The intersection coordinates can be written in matrix form as:

$$\mathbf{X} = (\mathbf{h} \quad \mathbf{m}) \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \quad (1.1.2.3)$$

- Degenerate Conic Condition: A quadratic form  $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$  represents a pair of straight lines if and only if the augmented matrix is singular:

$$\det \begin{pmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{pmatrix} = 0 \quad (1.1.2.4)$$

- Pencil of Conics and Common Chord: The intersection of two conics with parameters  $(\mathbf{V}_1, \mathbf{u}_1, f_1)$  and  $(\mathbf{V}_2, \mathbf{u}_2, f_2)$  is given by the pencil:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.1.2.5)$$

The value of  $\mu$  corresponding to the degenerate conic (the radical axis / common chord line) is found from:

$$\det \begin{pmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{pmatrix} = 0 \quad (1.1.2.6)$$

### 1.1.3 Laplace Transform

- First Derivative Property:

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.1.3.1)$$

- Transform of a Constant:

$$1 \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (1.1.3.2)$$

- First Frequency Shift / Exponential Inverse:

$$e^{at} \xleftrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.1.3.3)$$

#### 1.1.4 Euler's Method of Forward Differences

- For a first-order differential equation  $\frac{dy}{dt} = f(t, y)$  with step size  $h$ :

$$y_{n+1} = y_n + hf(t_n, y_n) \quad (1.1.4.1)$$

#### 1.1.5 Gradient Descent Optimization

- First-Order Parameter Update: For a differentiable function  $f(x)$ , the iterative parameter update with learning rate  $\eta > 0$  is given by:

$$x_{k+1} = x_k - \eta f'(x_k) \quad (1.1.5.1)$$

- Second Derivative Test for Convexity:

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \quad (1.1.5.2)$$

A strictly convex function on an open interval possesses at most one local minimum, which is also its unique global minimum.

#### 1.1.6 Minimization via Completing the Square

- For any  $x > 0$  and constant  $a > 0$ :

$$x + \frac{a^2}{x} = \left( \sqrt{x} - \frac{a}{\sqrt{x}} \right)^2 + 2a \geq 2a \quad (1.1.6.1)$$

The equality holds if and only if  $\sqrt{x} = \frac{a}{\sqrt{x}} \implies x = a$ , giving the minimum value  $2a$ .

#### 1.1.7 Newton-Raphson Method

- First-Order Recurrence (Difference Equation): For a real-valued differentiable function  $f(x)$ , the root is iteratively approximated using its first-order Taylor series expansion about the current estimate  $x_n$ :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.7.1)$$

#### 1.1.8 System of Linear Equations and Rouché–Capelli Theorem

- A system of  $m$  linear equations in  $n$  variables can be represented in matrix form as:

$$\mathbf{Ax} = \mathbf{b} \quad (1.1.8.1)$$

where  $\mathbf{A} \in \mathbb{R}^{m \times n}$  is the coefficient matrix,  $\mathbf{x} \in \mathbb{R}^{n \times 1}$  is the vector of unknowns, and  $\mathbf{b} \in \mathbb{R}^{m \times 1}$  is the constant vector.

- The solvability of the system is governed by the ranks of the coefficient matrix  $\mathbf{A}$  and the augmented matrix  $[\mathbf{A} \mid \mathbf{b}]$ :
  - a) Inconsistent System (No Solution): A linear system has no solution if and only if:

$$\text{rank}(\mathbf{A}) < \text{rank}([\mathbf{A} \mid \mathbf{b}]) \quad (1.1.8.2)$$

Geometrically, this corresponds to distinct parallel lines that do not intersect.

- b) Consistent System (At Least One Solution): A linear system has at least one solution if and only if:

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.1.8.3)$$

- If  $r = n$  (where  $n$  is the number of variables), there exists a unique solution.
- If  $r < n$ , there exist infinitely many solutions with  $n - r$  free variables (coincident lines).

### 1.1.9 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let  $X_1, X_2 \in \{1, 2, 3, 4, 5, 6\}$  be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF):

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.9.1)$$

- Let  $X = X_1 + X_2$  denote their sum. The PMF of  $X$  is given by the discrete convolution of  $p_{X_1}(n)$  and  $p_{X_2}(n)$ :

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n - k \mid X_2 = k) p_{X_2}(k) \quad (1.1.9.2)$$

$$= \sum_k p_{X_1}(n - k) p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \quad (1.1.9.3)$$

- Substituting (1.1.9.1) into (1.1.9.3) yields the triangular PMF:

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \leq n \leq 7 \\ \frac{13-n}{36}, & 7 < n \leq 12 \\ 0, & n > 12 \end{cases} \quad (1.1.9.4)$$

### 1.1.10 Unit Step Response of an $RC$ Circuit and $s$ -Domain Analysis

- Time-Domain Circuit Equation: Consider a series  $RC$  circuit excited by a step input  $v_i(t) = V_{dc}u(t)$  where  $V_{dc}$  is a constant DC voltage source, and  $u(t)$  is

step input function defined as:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (1.1.10.1)$$

Starting at  $t = 0$ . The current through the capacitor is:

$$i(t) = \frac{dq}{dt} = C \frac{dV_{out}}{dt} \quad (1.1.10.2)$$

Applying Kirchhoff's Voltage Law (KVL) around the loop:

$$Ri(t) + V_{out}(t) = V_{dc} \implies RC \frac{dV_{out}}{dt} + V_{out}(t) = V_{dc} \quad (1.1.10.3)$$

- *s*-Domain Representation: Transforming the circuit to the Laplace domain, the step source becomes  $V_{in}(s) = \frac{V_{dc}}{s}$ , the resistor impedance is  $R$ , and the capacitor impedance is  $\frac{1}{sC}$ . The equivalent impedance is:

$$R_{eq}(s) = R + \frac{1}{sC} = \frac{RsC + 1}{sC} \quad (1.1.10.4)$$

The loop current in the *s*-domain is:

$$I(s) = \frac{V_{in}(s)}{R_{eq}(s)} = \frac{\frac{V_{dc}}{s}}{\frac{RsC + 1}{sC}} = \frac{V_{dc}C}{RsC + 1} \quad (1.1.10.5)$$

The output voltage across the capacitor is obtained as:

$$V_{out}(s) = I(s) \left( \frac{1}{sC} \right) = \frac{V_{dc}}{s(RCs + 1)} \quad (1.1.10.6)$$

$$= \frac{V_{dc}}{RC} \left[ \frac{1}{s(s + \frac{1}{RC})} \right] = V_{dc} \left[ \frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right] \quad (1.1.10.7)$$

Taking the inverse Laplace transform yields:

$$V_{out}(t) = V_{dc} \left( 1 - e^{-\frac{t}{RC}} \right) u(t) \quad (1.1.10.8)$$

#### 1.1.11 Standard First-Order Differential Equation Comparison

- A general first-order dynamic system with time constant  $\tau$  subjected to a unit step input is described by:

$$\tau \frac{dy}{dt} + y(t) = y_{ss} \quad (1.1.11.1)$$

where  $\tau$  is the time constant and  $y_{ss}$  is the steady-state response.

- Comparing (1.1.11.1) with the  $RC$  circuit in (1.1.10.3):

$$y(t) \longleftrightarrow V_{out}(t), \quad \tau \longleftrightarrow RC, \quad y_{ss} \longleftrightarrow V_{dc} \quad (1.1.11.2)$$

Transforming (1.1.11.1) into the  $s$ -domain with zero initial conditions  $y(0) = 0$ :

$$\tau s Y(s) + Y(s) = \frac{y_{ss}}{s} \implies Y(s) = \frac{y_{ss}}{s(\tau s + 1)} = y_{ss} \left[ \frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} \right] \quad (1.1.11.3)$$

Taking the inverse Laplace transform directly gives:

$$y(t) = y_{ss} \left( 1 - e^{-\frac{t}{\tau}} \right) u(t) \quad (1.1.11.4)$$

#### 1.1.12 Transfer Function of a First-Order Process

- The transfer function of a linear time-invariant system is defined as the ratio of the Laplace transform of the output  $Y(s)$  to the Laplace transform of the input  $F(s)$ , evaluated under the assumption that all initial conditions are zero.

$$H(s) = \frac{Y(s)}{F(s)} \quad (1.1.12.1)$$

#### 1.1.13 Identity Condition for Linear Combinations

- For any independent variables or functions  $x$  and  $y$ , if the relation

$$ax + by = 0 \quad (1.1.13.1)$$

holds identically for all  $x$  and  $y$ , then the coefficients must vanish simultaneously:

$$a = 0 \quad \text{and} \quad b = 0 \quad (1.1.13.2)$$

- Similarly, if an expression  $C_1 f_1(t) + C_2 f_2(t) = 0$  holds for arbitrary constants  $C_1$  and  $C_2$ , each coefficient must vanish independently:

$$f_1(t) = 0 \quad \text{and} \quad f_2(t) = 0 \quad (1.1.13.3)$$

For distinct exponential functions  $e^{r_1 t}$  and  $e^{r_2 t}$  ( $r_1 \neq r_2$ ), this implies linear independence across all  $t$ .

#### 1.1.14 Continuity and Differentiability in Functions

- Continuity at a Point: A real-valued function  $f(x)$  is continuous at  $x = x_0$  if and only if the left-hand limit, the right-hand limit, and the value of the function coincide:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = f(x_0) \quad (1.1.14.1)$$

- Differentiability and Slope Matching: A function  $f(x)$  is differentiable at  $x = x_0$  if the left-hand derivative equals the right-hand derivative:

$$f'(x_0^-) = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0^+) \quad (1.1.14.2)$$

For piecewise functions defined by differentiable curves on each side of the

boundary  $x_0$ , this requires matching boundary slopes:

$$f'(x_0^-) = f'(x_0^+) \quad (1.1.14.3)$$

- **Differentiability Implies Continuity:** If  $f(x)$  is differentiable at  $x = x_0$ , it must necessarily be continuous at  $x = x_0$ :

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = \lim_{x \rightarrow x_0} \left[ \frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \quad (1.1.14.4)$$

Hence, continuity at  $x = x_0$  is a necessary prerequisite for differentiability.

- 1.1.15 **Piecewise Boundary Conditions at  $x = a$ :** Consider a function  $f(x)$  defined by differentiable branches  $f_1(x)$  and  $f_2(x)$  across a boundary point  $x = a$ :

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \quad (1.1.15.1)$$

For  $f(x)$  to be continuous at  $x = a$ , the one-sided limits must equal the boundary value:

$$\lim_{x \rightarrow a^-} f_1(x) = \lim_{x \rightarrow a^+} f_2(x) \implies f_1(a) = f_2(a) \quad (1.1.15.2)$$

For  $f(x)$  to be differentiable at  $x = a$ , the one-sided derivatives at the boundary must coincide:

$$f'_1(a) = f'_2(a) \quad (1.1.15.3)$$

- 1.1.16 **Representation of Linear Balance Systems as Matrix Equations**

- A general system of  $m$  linear equations in  $n$  variables  $x_1, x_2, \dots, x_n$ :

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, m \quad (1.1.16.1)$$

can be expressed in matrix-vector form:

$$\mathbf{Ax} = \mathbf{b} \quad (1.1.16.2)$$

where  $\mathbf{A} \in \mathbb{R}^{m \times n}$  is the coefficient matrix,  $\mathbf{x} \in \mathbb{R}^n$  is the state vector, and  $\mathbf{b} \in \mathbb{R}^m$  is the constant vector:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad (1.1.16.3)$$

- In multi-component conservation and stoichiometric balances, the conservation of elemental species across reactant and product streams is represented by:

$$\mathbf{A}_{\text{reactants}} \mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{products}} \mathbf{x}_{\text{products}} \quad (1.1.16.4)$$

Combining both sides into a single stoichiometric matrix yields the homogeneous

system:

$$[\mathbf{A}_{\text{reactants}} - \mathbf{A}_{\text{products}}] \mathbf{x} = \mathbf{0} \quad (1.1.16.5)$$

When boundary parameters or specific yields fix certain degrees of freedom (such as basis mole values and yield coefficients), the homogeneous system reduces to a deterministic non-homogeneous matrix equation  $\mathbf{A}\mathbf{x} = \mathbf{b}$  that can be solved directly via row operations or symbolic matrix inversion.

#### 1.1.17 Solution of First-Order Recurrence Relations via Z-Transform

- The bilateral Z-transform of a discrete-time sequence  $x[n]$  is defined as:

$$X(z) = \mathcal{Z}[x[n]] = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \quad (1.1.17.1)$$

over the Region of Convergence (ROC) where the series converges absolutely.

- Discrete Unit Step Sequence: The discrete unit step function  $u[n]$  is defined as:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.1.17.2)$$

Its forward Z-transform is given by:

$$\mathcal{Z}[u[n]] = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \quad (1.1.17.3)$$

- Time-Shift Property: Shifting a sequence by an integer delay  $k$  corresponds to multiplication by  $z^{-k}$  in the Z-domain:

$$\mathcal{Z}[x[n - k]] = z^{-k} X(z) \quad (1.1.17.4)$$

Consequently, delaying the discrete unit step by one sample yields:

$$\mathcal{Z}[u[n - 1]] = \frac{z^{-1}}{1 - z^{-1}}, \quad |z| > 1 \quad (1.1.17.5)$$

- Standard Inverse Transform Pair: For any scalar constant  $\alpha$ :

$$\mathcal{Z}^{-1} \left[ \frac{1}{1 - \alpha z^{-1}} \right] = \alpha^n u[n], \quad |z| > |\alpha| \quad (1.1.17.6)$$

#### 1.1.18 Relation Between Eigenvalues and Trace of a Matrix

- Let  $\mathbf{A} \in \mathbb{R}^{n \times n}$  be a square matrix with eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ . The characteristic polynomial of  $\mathbf{A}$  is defined by:

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (1.1.18.1)$$

- Factoring the polynomial in terms of its  $n$  characteristic roots:

$$P(\lambda) = \prod_{i=1}^n (\lambda - \lambda_i) = \lambda^n - \left( \sum_{i=1}^n \lambda_i \right) \lambda^{n-1} + \dots + (-1)^n \prod_{i=1}^n \lambda_i \quad (1.1.18.2)$$

- Evaluating the determinant  $|\lambda \mathbf{I} - \mathbf{A}|$  using cofactor expansion, the only term that contributes to  $\lambda^n$  and  $\lambda^{n-1}$  is the product of the principal diagonal entries:

$$\prod_{i=1}^n (\lambda - a_{ii}) = \lambda^n - \left( \sum_{i=1}^n a_{ii} \right) \lambda^{n-1} + \dots \quad (1.1.18.3)$$

where  $a_{ii}$  denotes the diagonal elements of  $\mathbf{A}$ .

- Equating the coefficient of  $\lambda^{n-1}$  from (1.1.18.2) and (1.1.18.3):

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.1.18.4)$$

Hence, the sum of all eigenvalues of a square matrix is always equal to its trace.

### B. Problems

1.2.1 The sum of the following infinite series is:

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots \quad (1.2.1.1)$$

(GATE BT 2025)

a)  $\pi$

c)  $e - 1$

b)  $1 + e$

d)  $e$

**Solution:** From the Maclaurin series expansion of  $e^x$  given in (1.1.1.1):

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad (1.2.1.2)$$

Evaluating the expression at  $x = 1$ :

$$e^1 = 1 + \frac{1^1}{1!} + \frac{1^2}{2!} + \frac{1^3}{3!} + \frac{1^4}{4!} + \dots \quad (1.2.1.3)$$

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \quad (\text{since } 1^n = 1) \quad (1.2.1.4)$$

Using the original equation:

$$\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots = \sum_{n=1}^{\infty} \frac{1}{n!} \quad (1.2.1.5)$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{1}{n!} - 1 = e - 1 \quad (1.2.1.6)$$

Hence, option (C) is correct.

The following C code verifies the series sum numerically:

codes/btq3/btq3.c

1.2.2 A circle with center at  $(x, y) = (0.5, 0)$  and radius  $r_1 = 0.5$  intersects with another circle with center at  $(x, y) = (1, 1)$  and radius  $r_2 = 1$  at two points. One of the points of intersection  $(x, y)$  is: (GATE BT 2025)

a)  $(0, 0)$

c)  $(0.5, 0.5)$

b)  $(0.2, 0.4)$

d)  $(1, 2)$

**Solution:** The matrix parameters for the two circles are formulated from their center

vectors  $\mathbf{c}_i$  and radii  $r_i$ :

$$\text{Circle 1: } \mathbf{c}_1 = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix}, r_1 = 0.5 \implies \mathbf{V}_1 = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_1 = -\mathbf{c}_1 = \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} \quad (1.2.2.1)$$

$$f_1 = \|\mathbf{c}_1\|^2 - r_1^2 = 0.5^2 - 0.5^2 = 0 \quad (1.2.2.2)$$

$$\text{Circle 2: } \mathbf{c}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, r_2 = 1 \implies \mathbf{V}_2 = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u}_2 = -\mathbf{c}_2 = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (1.2.2.3)$$

$$f_2 = \|\mathbf{c}_2\|^2 - r_2^2 = (1^2 + 1^2) - 1^2 = 1 \quad (1.2.2.4)$$

Using the pencil of conics from (1.1.2.5), the condition for a degenerate conic from (1.1.2.6) gives:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.2.2.5)$$

Substituting the parameter matrices:

$$\begin{vmatrix} 1 + \mu & 0 & -0.5 - \mu \\ 0 & 1 + \mu & -\mu \\ -0.5 - \mu & -\mu & \mu \end{vmatrix} = 0 \quad (1.2.2.6)$$

Expanding the determinant:

$$(+\mu) [(1 + \mu) \mu - \mu^2] - (-0.5 - \mu) [0 - (1 + \mu) (-0.5 - \mu)] = 0 \quad (1.2.2.7)$$

$$(1 + \mu) \mu - (1 + \mu) (\mu + 0.5)^2 = 0 \quad (1.2.2.8)$$

$$(1 + \mu) \left( -\mu^2 - \frac{1}{4} \right) = 0 \quad (1.2.2.9)$$

Since  $\mu^2 + \frac{1}{4} = 0$  yields imaginary roots, the only real root is:

$$\mu = -1 \quad (1.2.2.10)$$

Substituting  $\mu = -1$  into (1.1.2.5) eliminates the second-degree terms ( $\mathbf{V}_1 - \mathbf{V}_2 = \mathbf{0}$ ), yielding the common chord (radical axis):

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.2.2.11)$$

$$2 \left( \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ -1 \end{pmatrix} \right)^\top \mathbf{x} + (0 - 1) = 0 \quad (1.2.2.12)$$

$$2(0.5 \ 1) \mathbf{x} - 1 = 0 \implies (1 \ 2) \mathbf{x} = 1 \quad (1.2.2.13)$$

Thus, the equation of the line passing through the intersection points is  $x + 2y = 1$ .

The normal vector of this line is  $\mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ . A direction vector  $\mathbf{m}$  orthogonal to  $\mathbf{n}$

$(\mathbf{m}^\top \mathbf{n} = 0)$  and a point  $\mathbf{h}$  on the line are:

$$\mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.2.2.14)$$

Expressing the line in parametric form  $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$  and substituting into Circle 1 using (1.1.2.2):

$$\mathbf{m}^\top \mathbf{V}_1 \mathbf{m} = (2 \quad -1) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 5 \quad (1.2.2.15)$$

$$\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \quad (1.2.2.16)$$

$$\mathbf{m}^\top (\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1) = (2 \quad -1) \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} = 1 \quad (1.2.2.17)$$

$$g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}_1 \mathbf{h} + 2\mathbf{u}_1^\top \mathbf{h} + f_1 = 1 - 1 + 0 = 0 \quad (1.2.2.18)$$

Evaluating the parameter roots  $\kappa$ :

$$\kappa = \frac{-1 \pm \sqrt{1^2 - 0}}{5} = \frac{-1 \pm 1}{5} \implies \kappa_1 = 0, \quad \kappa_2 = -\frac{2}{5} \quad (1.2.2.19)$$

The points of intersection are:

$$\mathbf{x}_1 = \mathbf{h} + 0\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1.2.2.20)$$

$$\mathbf{x}_2 = \mathbf{h} - \frac{2}{5}\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \frac{2}{5} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix} \quad (1.2.2.21)$$

Hence, one of the points of intersection is  $(0.2, 0.4)$ , and option (B) is correct.

|                              |
|------------------------------|
| codes/btq9/circleProblem1.py |
| figs/circleintersect.pdf     |

$$\frac{dy}{dt} = \lambda(y + 2) \quad (1.2.3.1)$$

a)  $\ln 2$   
b)  $\ln 3$

$$\frac{dy}{dt} - \lambda y = 2\lambda \quad (1.2.3.2)$$
$$sY(s) - y(0) - \lambda Y(s) = \frac{2\lambda}{s} \quad (1.2.3.3)$$

Substituting the initial condition  $y(0) = 1$ :

$$(s - \lambda) Y(s) - 1 = \frac{2\lambda}{s} \quad (1.2.3.4)$$

$$(s - \lambda) Y(s) = 1 + \frac{2\lambda}{s} = \frac{s + 2\lambda}{s} \quad (1.2.3.5)$$

$$Y(s) = \frac{s + 2\lambda}{s(s - \lambda)} \quad (1.2.3.6)$$

Decomposing into partial fractions:

$$\frac{s + 2\lambda}{s(s - \lambda)} = \frac{A}{s} + \frac{B}{s - \lambda} \implies s + 2\lambda = A(s - \lambda) + Bs \quad (1.2.3.7)$$

1.2.4 Evaluating the coefficients:

$$\text{At } s = 0 : \quad 2\lambda = A(-\lambda) \implies A = -2 \quad (1.2.4.1)$$

$$\text{At } s = \lambda : \quad 3\lambda = B(\lambda) \implies B = 3 \quad (1.2.4.2)$$

Substituting  $A$  and  $B$  back into  $Y(s)$  and applying the inverse Laplace transform using (1.1.3.2) and (1.1.3.3):

$$Y(s) = -\frac{2}{s} + \frac{3}{s - \lambda} \quad (1.2.4.3)$$

$$y(t) = 3e^{\lambda t} - 2 \quad (1.2.4.4)$$

Applying the boundary condition  $y(1) = 4$ :

$$4 = 3e^{\lambda(1)} - 2 \quad (1.2.4.5)$$

$$6 = 3e^{\lambda} \quad (1.2.4.6)$$

$$e^{\lambda} = 2 \implies \lambda = \ln 2 \quad (1.2.4.7)$$

Hence, option (A) is correct.

Numerical Simulation (Euler's Method):

Applying the recurrence relation from (1.1.4.1) with step size  $h = 0.01$  and  $\lambda = \ln 2$ :

$$y_{n+1} = y_n + 0.01 [\ln(2)(y_n + 2)], \quad y_0 = 1 \quad (1.2.4.8)$$

The C code implementation is as follows:

codes/btq13/bac.c



For all  $x \in (0, \infty)$ ,  $f''(x) > 0$ . Thus,  $f(x)$  is strictly convex, and  $x = 2$  corresponds to a unique global minimum.

Evaluating the function at  $x = 2$ :

$$f(2) = 2 + \frac{4}{2} = 4 \quad (1.2.5.6)$$

Hence, option (D) is correct.

Numerical Simulation (Gradient Descent):

Applying the first-order update relation from (1.1.5.1) with learning rate  $\eta = 0.3$ :

$$x_{k+1} = x_k - 0.3 \left( 1 - \frac{4}{x_k^2} \right) \quad (1.2.5.7)$$

Starting from the initial guess  $x_0 = 5$ , the sequence converges to  $x^* = 2$ , yielding  $f(x^*) = 4$ .

1.2.6 Alternative Method (Completing the Square): For  $x > 0$ , the function  $f(x)$  can be expressed in terms of square roots and rewritten as a perfect square:

$$f(x) = x + \frac{4}{x} \quad (1.2.6.1)$$

$$= (\sqrt{x})^2 - 2(\sqrt{x}) \left( \frac{2}{\sqrt{x}} \right) + \left( \frac{2}{\sqrt{x}} \right)^2 + 2(\sqrt{x}) \left( \frac{2}{\sqrt{x}} \right) \quad (1.2.6.2)$$

$$= \left( \sqrt{x} - \frac{2}{\sqrt{x}} \right)^2 + 4 \quad (1.2.6.3)$$

Since the square of any real number is non-negative:

$$\left( \sqrt{x} - \frac{2}{\sqrt{x}} \right)^2 \geq 0 \quad \forall x > 0 \quad (1.2.6.4)$$

with equality holding if and only if:

$$\sqrt{x} - \frac{2}{\sqrt{x}} = 0 \implies \sqrt{x} = \frac{2}{\sqrt{x}} \implies x = 2 \quad (1.2.6.5)$$

Therefore:

$$f(x) \geq 4 \quad \forall x > 0 \quad (1.2.6.6)$$

The minimum value is 4, attained at  $x = 2$ , which confirms option (D).

The C code implementing iterative gradient descent is:

```
codes/btq14/min.c
```

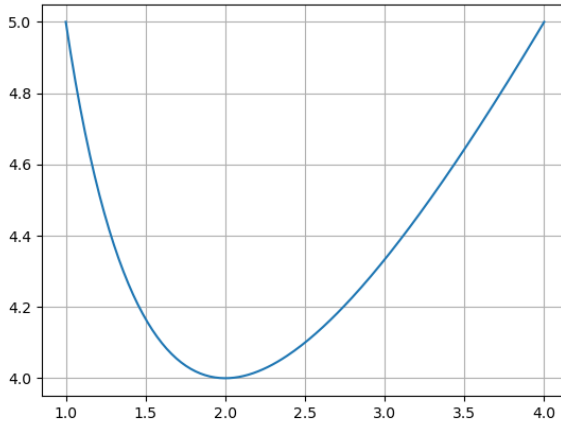


Fig. 1.2.6.1. Plot of  $f(x) = x + \frac{4}{x}$  indicating the minimum at  $(2, 4)$

codes/btq14/min.py

figs/min.png

1.2.7 Consider a nonlinear algebraic equation,  $e^x - 2 = 0$ . Using the Newton-Raphson method, with the initial guess of  $x_0 = 1$ , the approximated value of the root of the equation after one iteration is \_\_\_\_\_ (round off to two decimal places). (GATE BT 2025)

**Solution:** The given objective function and its first derivative are:

$$f(x) = e^x - 2 \quad (1.2.7.1)$$

$$f'(x) = e^x \quad (1.2.7.2)$$

Substituting  $f(x)$  and  $f'(x)$  into the Newton-Raphson recurrence relation from (1.1.7.1):

$$x_{n+1} = x_n - \frac{e^{x_n} - 2}{e^{x_n}} \quad (1.2.7.3)$$

$$= x_n - \left(1 - \frac{2}{e^{x_n}}\right) \quad (1.2.7.4)$$

$$= x_n - 1 + 2e^{-x_n} \quad (1.2.7.5)$$

Given the initial guess  $x_0 = 1$ , the first iteration ( $n = 0$ ) gives:

$$x_1 = x_0 - 1 + 2e^{-x_0} \quad (1.2.7.6)$$

$$= 1 - 1 + 2e^{-1} \quad (1.2.7.7)$$

$$= \frac{2}{e} \approx \frac{2}{2.71828} \approx 0.73576 \approx 0.74 \quad (1.2.7.8)$$

Subsequent Iterations (Convergence to  $\ln 2$ ):

$$x_2 = x_1 - 1 + 2e^{-x_1} \approx 0.694042 \quad (1.2.7.9)$$

$$x_3 = x_2 - 1 + 2e^{-x_2} \approx 0.693148 \quad (1.2.7.10)$$

$$x_4 = x_3 - 1 + 2e^{-x_3} \approx 0.693147 \approx \ln 2 \quad (1.2.7.11)$$

The exact mathematical solution is  $e^x = 2 \implies x = \ln 2 \approx 0.693147$ . Thus, the approximated value of the root after one iteration is 0.74.

The C Code implementing Newton-Raphson recurrence is as follows:

codes/btq33/raph.c

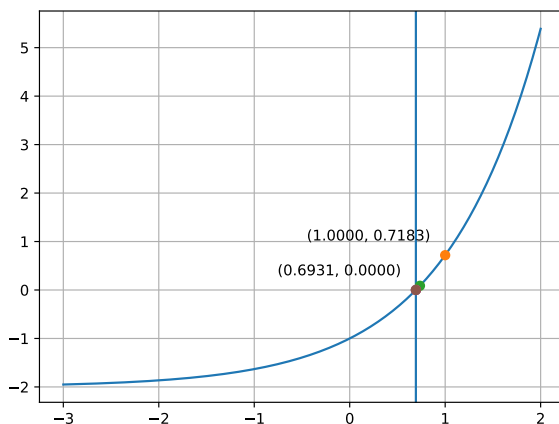


Fig. 1.2.7.1. Convergence of the Newton-Raphson method to  $x = \ln 2$

codes/btq33/raph.py

1.2.8 The value of  $k$ , for which the linear equations  $2x + 3y = 6$  and  $4x + 6y = 3k$  have at least one solution, is \_\_\_\_\_ (answer in integer). (GATE BT 2025)

**Solution:** The given system of linear equations is:

$$2x + 3y = 6 \quad (1.2.8.1)$$

$$4x + 6y = 3k \quad (1.2.8.2)$$

Expressing in matrix form  $\mathbf{Ax} = \mathbf{b}$ :

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 3k \end{pmatrix} \quad (1.2.8.3)$$

The augmented matrix is formulated as:

$$(\mathbf{A} \ \mathbf{b} \ |) = \left( \begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array} \right) \quad (1.2.8.4)$$

Applying Gaussian elimination using the elementary row operation  $R_2 \rightarrow R_2 - 2R_1$ :

$$\left( \begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left( \begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3k - 12 \end{array} \right) \quad (1.2.8.5)$$

The coefficient matrix  $\mathbf{A}$  reduces to:

$$\mathbf{A} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & 3 \\ 0 & 0 \end{pmatrix} \quad (1.2.8.6)$$

Since there is only one non-zero row in echelon form,  $\text{rank}(\mathbf{A}) = 1$ .

From the Rouché–Capelli theorem in (1.1.8.3), for the system to have at least one solution, it must be consistent:

$$\text{rank}((\mathbf{A} \ \mathbf{b} \ |)) = \text{rank}(\mathbf{A}) = 1 \quad (1.2.8.7)$$

This requires the second row of the augmented matrix to be entirely zeros:

$$3k - 12 = 0 \quad (1.2.8.8)$$

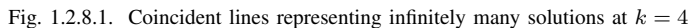
$$3k = 12 \implies k = 4 \quad (1.2.8.9)$$

For  $k = 4$ ,  $\text{rank}(\mathbf{A}) = \text{rank}((\mathbf{A} \ \mathbf{b} \ |)) = 1 < 2$  (number of variables  $n = 2$ ), yielding infinitely many solutions along the coincident line  $2x + 3y = 6$ .

Hence, the integer value of  $k$  is 4.

The Python Code showing the implementation for a given value of  $k$  and comparing solutions is as follows:

codes/btq34/bt34.py



codes/btq35/dice.py



The instrument reaches 95% of its steady-state reading when  $y(t_{95}) = 0.95y_{ss}$ :

$$y_{ss} \left(1 - e^{-\frac{t_{95}}{\tau}}\right) = 0.95y_{ss} \quad (1.2.10.4)$$

$$1 - e^{-\frac{t_{95}}{\tau}} = 0.95 \quad (1.2.10.5)$$

$$e^{-\frac{t_{95}}{\tau}} = 0.05 = \frac{1}{20} \quad (1.2.10.6)$$

Taking the natural logarithm on both sides:

$$-\frac{t_{95}}{\tau} = \ln(0.05) = -\ln(20) \quad (1.2.10.7)$$

$$t_{95} = \tau \ln(20) \quad (1.2.10.8)$$

Substituting the given time constant  $\tau = 40$  s:

$$t_{95} = 40 \ln(20) \quad (1.2.10.9)$$

$$\approx 40 \times 2.99573 \approx 119.83 \text{ s} \quad (1.2.10.10)$$

Rounding off to the nearest integer gives 120 seconds. Hence, option (C) is correct.

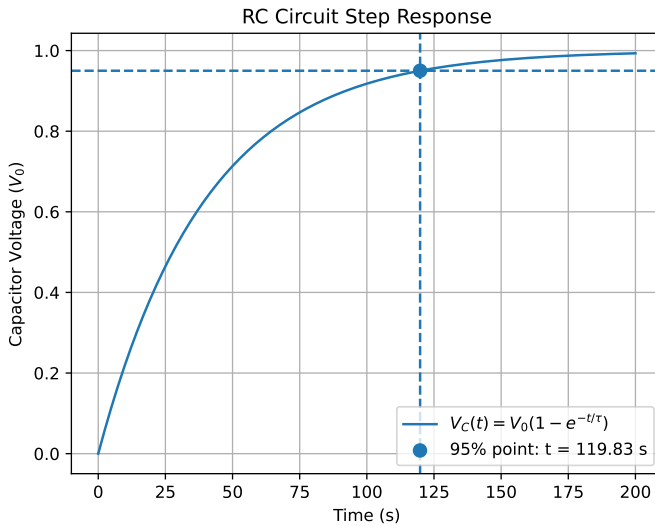


Fig. 1.2.10.1. Step response of the first-order system with  $\tau = 40$  s

codes/btq42/plot.py

figs/rc-step-response.pdf

1.2.11 The output  $y(t)$  of a first-order process is governed by the following differential equation

$$\tau_p \frac{dy}{dt} + y = K_p f(t) \quad (1.2.11.1)$$

where  $\tau_p$  is a non-zero time constant,  $K_p$  is the gain and  $f(t)$  is the input with  $f(0) = 0$ .

Assume  $y(0) = 0$ . The transfer function for this process is (consider  $s$  as the independent variable in the Laplace domain) (GATE BT 2025)

- |                               |                              |
|-------------------------------|------------------------------|
| a) $\frac{K_p}{\tau_p s + 1}$ | c) $\frac{\tau_p}{K_p(s+1)}$ |
| b) $\frac{\tau_p}{K_p s + 1}$ | d) $\frac{K_p}{\tau_p(s+1)}$ |

**Solution:** The given first-order differential equation governing the process is:

$$\tau_p \frac{dy(t)}{dt} + y(t) = K_p f(t) \quad (1.2.11.2)$$

Taking the unilateral Laplace transform on both sides using (1.1.3.1):

$$\tau_p [sY(s) - y(0)] + Y(s) = K_p F(s) \quad (1.2.11.3)$$

Applying the given initial condition  $y(0) = 0$ :

$$\tau_p sY(s) + Y(s) = K_p F(s) \quad (1.2.11.4)$$

$$Y(s)(\tau_p s + 1) = K_p F(s) \quad (1.2.11.5)$$

The transfer function is the ratio of the output  $Y(s)$  to the input  $F(s)$ :

$$\frac{Y(s)}{F(s)} = \frac{K_p}{\tau_p s + 1} \quad (1.2.11.6)$$

Hence, option (A) is correct.

1.2.12 Let  $m$  and  $n$  be fixed real numbers. If the function  $y(t) = C_1 e^t + C_2 e^{-t}$  is a solution of

$$\frac{d^2 y}{dt^2} + m \frac{dy}{dt} + ny = 0 \quad (1.2.12.1)$$

for any constants  $C_1$  and  $C_2$ , then  $m + n$  is equal to (GATE BT 2025)

- |       |      |
|-------|------|
| a) -2 | c) 0 |
| b) -1 | d) 1 |

**Solution:** The given function is:

$$y(t) = C_1 e^t + C_2 e^{-t} \quad (1.2.12.2)$$

Computing the first and second derivatives with respect to  $t$ :

$$\frac{dy}{dt} = C_1 e^t - C_2 e^{-t} \quad (1.2.12.3)$$

$$\frac{d^2 y}{dt^2} = C_1 e^t + C_2 e^{-t} \quad (1.2.12.4)$$

Substituting  $y$ ,  $\frac{dy}{dt}$ , and  $\frac{d^2 y}{dt^2}$  into the differential equation:

$$\frac{d^2 y}{dt^2} + m \frac{dy}{dt} + ny = 0 \quad (1.2.12.5)$$

$$(C_1 e^t + C_2 e^{-t}) + m(C_1 e^t - C_2 e^{-t}) + n(C_1 e^t + C_2 e^{-t}) = 0 \quad (1.2.12.6)$$

Grouping the terms containing  $C_1$  and  $C_2$ :

$$C_1 e^t (1 + m + n) + C_2 e^{-t} (1 - m + n) = 0 \quad (1.2.12.7)$$

Using (1.1.13.1) and (1.1.13.2), since this equation must hold for any constants  $C_1$  and  $C_2$ :

$$1 + m + n = 0 \quad (1.2.12.8)$$

$$1 - m + n = 0 \quad (1.2.12.9)$$

Adding (1.2.12.8) and (1.2.12.9):

$$2 + 2n = 0 \implies n = -1 \quad (1.2.12.10)$$

Subtracting (1.2.12.9) from (1.2.12.8):

$$2m = 0 \implies m = 0 \quad (1.2.12.11)$$

Evaluating  $m + n$ :

$$m + n = 0 + (-1) = -1 \quad (1.2.12.12)$$

Hence, option (B) is correct.

1.2.13 Consider the piecewise function

$$f(x) = \begin{cases} \sin(2x), & x > 0 \\ a + bx, & x \leq 0 \end{cases} \quad (1.2.13.1)$$

where  $a$  and  $b$  are real constants. If  $f(x)$  is differentiable at  $x = 0$ , then the value of  $a + b$  is (GATE BT 2025)

a) 0

c) 2

b) 1

d) -2

**Solution:** Since  $f(x)$  is differentiable at  $x = 0$ , from (1.1.14.4) it must also be continuous at  $x = 0$ . Applying the continuity condition from (1.1.14.1):

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0) \quad (1.2.13.2)$$

Evaluating the one-sided limits:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sin(2x) = \sin(0) = 0 \quad (1.2.13.3)$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (a + bx) = a + b(0) = a \quad (1.2.13.4)$$

Equating the limits yields:

$$a = 0 \quad (1.2.13.5)$$

Applying the differentiability condition and matching one-sided derivatives at  $x = 0$  using (1.1.14.3):

$$f'(0^+) = f'(0^-) \quad (1.2.13.6)$$

For  $x > 0$ :

$$f'(x) = \frac{d}{dx} (\sin(2x)) = 2 \cos(2x) \implies f'(0^+) = 2 \cos(0) = 2 \quad (1.2.13.7)$$

For  $x < 0$ :

$$f'(x) = \frac{d}{dx} (a + bx) = b \implies f'(0^-) = b \quad (1.2.13.8)$$

Equating the two slopes:

$$b = 2 \quad (1.2.13.9)$$

Evaluating  $a + b$ :

$$a + b = 0 + 2 = 2 \quad (1.2.13.10)$$

Hence, option (C) is correct.

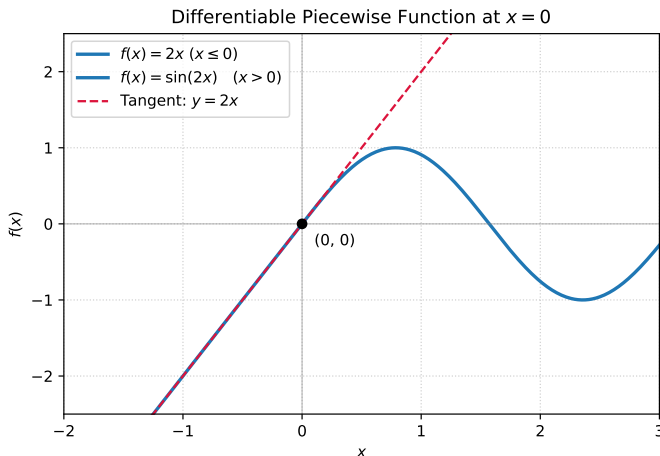
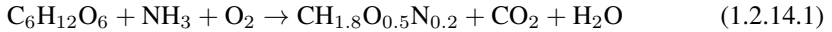


Fig. 1.2.13.1. Differentiable piecewise function and its tangent line  $y = 2x$  at  $x = 0$

codes/btq46/bt46.py

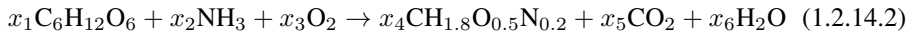
figs/piecewise-tangent-plot.pdf

- 1.2.14 In a fermentation process, each mole of glucose is converted to biomass ( $\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$ ), with a biomass yield coefficient of 0.4 C-mol/C-mol, according to the unbalanced equation given below.



The moles of oxygen consumption per mole of glucose consumed during fermentation is \_\_\_\_\_. (GATE BT 2025)

**Solution:** Let the balanced stoichiometric reaction be represented as:



where  $x_1, x_2, x_3, x_4, x_5, x_6$  are the respective stoichiometric coefficients.

Formulating the elemental conservation balances for C, N, H, and O using exact rational fractions:

$$\text{Carbon (C): } 6x_1 = x_4 + x_5 \quad (1.2.14.3)$$

$$\text{Nitrogen (N): } x_2 = \frac{1}{5}x_4 \quad (1.2.14.4)$$

$$\text{Hydrogen (H): } 12x_1 + 3x_2 = \frac{9}{5}x_4 + 2x_6 \quad (1.2.14.5)$$

$$\text{Oxygen (O): } 6x_1 + 2x_3 = \frac{1}{2}x_4 + 2x_5 + x_6 \quad (1.2.14.6)$$

Writing the elemental balances in matrix form  $\mathbf{A}_{\text{LHS}}\mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{RHS}}\mathbf{x}_{\text{products}}$ :

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 12 & 3 & 0 \\ 6 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{5} & 0 & 0 \\ 0 & 2 \\ \frac{1}{2} & 2 & 1 \end{pmatrix} \begin{pmatrix} x_4 \\ x_5 \\ x_6 \end{pmatrix} \quad (1.2.14.7)$$

Constructing the stoichiometric matrix  $\mathbf{A} = [\mathbf{A}_{\text{LHS}} \quad -\mathbf{A}_{\text{RHS}}]$ :

$$\begin{pmatrix} 6 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & 0 & 0 \\ 12 & 3 & 0 & -\frac{9}{5} & 0 & -2 \\ 6 & 0 & 2 & -\frac{1}{2} & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (1.2.14.8)$$

The biomass yield coefficient is 0.4 C-mol/C-mol =  $\frac{2}{5}$  C-mol/C-mol, which provides the relation:

$$x_4 - \frac{2}{5}(6x_1) = 0 \implies x_4 = \frac{12}{5}x_1 \quad (1.2.14.9)$$

Solving the system symbolically in terms of  $x_4$ :

$$x_1 = \frac{5}{12}x_4 \quad (1.2.14.10)$$

$$x_2 = \frac{1}{5}x_4 \quad (1.2.14.11)$$

$$x_3 = \frac{29}{20}x_4 \quad (1.2.14.12)$$

$$x_5 = \frac{3}{2}x_4 \quad (1.2.14.13)$$

$$x_6 = \frac{19}{10}x_4 \quad (1.2.14.14)$$

Evaluating with the sample numerical basis  $x_4 = 5.0$ :

$$x_1 = \frac{25}{12} \approx 2.0833 \quad (1.2.14.15)$$

$$x_2 = 1.0000 \quad (1.2.14.16)$$

$$x_3 = 7.2500 \quad (1.2.14.17)$$

$$x_4 = 5.0000 \quad (1.2.14.18)$$

$$x_5 = 7.5000 \quad (1.2.14.19)$$

$$x_6 = 9.5000 \quad (1.2.14.20)$$

The moles of oxygen consumed per mole of glucose consumed during fermentation is given by:

$$\frac{x_3}{x_1} = \frac{\frac{29}{20}x_4}{\frac{5}{12}x_4} = \frac{29}{20} \times \frac{12}{5} = \frac{87}{25} = 3.48 \quad (1.2.14.21)$$

or using the numerical evaluation:

$$\frac{x_3}{x_1} = \frac{7.2500}{2.0833} = 3.48 \quad (1.2.14.22)$$

Rounding off to two decimal places, the answer is 3.48.

The implementation in SymPy to find solutions is as follows:

codes/btq58/58.py

- 1.2.15 Let  $a_0 = 0$  and define  $a_n = \frac{1}{2}(1 + a_{n-1})$  for all positive integers  $n \geq 1$ . The least value of  $n$  for which  $|1 - a_n| < \frac{1}{2^{10}}$  is \_\_\_\_\_. (GATE BT 2025)

**Solution:** The given recurrence relation is:

$$a_n = \frac{1}{2}(1 + a_{n-1}), \quad n \geq 1, \quad a_0 = 0 \quad (1.2.15.1)$$

Extending the difference equation across all  $n \in \mathbb{Z}$  by causal multiplication with  $u[n]$  from (1.1.17.2):

$$a_n u[n] - \frac{1}{2} a_{n-1} u[n-1] = \frac{1}{2} u[n-1] \quad (1.2.15.2)$$

Let  $A(z) = \mathcal{Z}[a_n u[n]]$ . Applying the time-shift property (1.1.17.4) and the delayed step transform (1.1.17.5):

$$A(z) - \frac{1}{2}z^{-1}A(z) = \frac{1}{2}\mathcal{Z}[u[n-1]] \quad (1.2.15.3)$$

$$A(z) \left(1 - \frac{1}{2}z^{-1}\right) = \frac{\frac{1}{2}z^{-1}}{1 - z^{-1}} \quad (1.2.15.4)$$

Solving algebraically for  $A(z)$ :

$$A(z) = \frac{\frac{1}{2}z^{-1}}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})}, \quad |z| > 1 \quad (1.2.15.5)$$

Performing partial fraction decomposition with  $w = z^{-1}$ :

$$\frac{\frac{1}{2}w}{(1-w)(1-\frac{1}{2}w)} = \frac{C_1}{1-w} + \frac{C_2}{1-\frac{1}{2}w} \quad (1.2.15.6)$$

where

$$C_1 = \left. \frac{\frac{1}{2}w}{1-\frac{1}{2}w} \right|_{w=1} = \frac{1/2}{1/2} = 1 \quad (1.2.15.7)$$

$$C_2 = \left. \frac{\frac{1}{2}w}{1-w} \right|_{w=2} = \frac{1}{1-2} = -1 \quad (1.2.15.8)$$

Substituting back into  $A(z)$ :

$$A(z) = \frac{1}{1-z^{-1}} - \frac{1}{1-\frac{1}{2}z^{-1}} \quad (1.2.15.9)$$

Applying the inverse Z-transform using (1.1.17.3) and (1.1.17.6):

$$a_n = \left[1 - \left(\frac{1}{2}\right)^n\right] u[n] = 1 - 2^{-n}, \quad n \geq 0 \quad (1.2.15.10)$$

Applying the stopping threshold condition  $|1 - a_n| < \frac{1}{2^{10}}$ :

$$|1 - (1 - 2^{-n})| < \frac{1}{2^{10}} \quad (1.2.15.11)$$

$$2^{-n} < 2^{-10} \quad (1.2.15.12)$$

$$2^n > 2^{10} \implies n > 10 \quad (1.2.15.13)$$

Since  $n$  must be an integer, the minimal integer satisfying the strict inequality is:

$$n = 11 \quad (1.2.15.14)$$

The implementation in C to simulate the recurrence and verify the threshold is as follows:

codes/btq59/ztransform.c

1.2.16 Let  $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix}$ . If the eigenvalues of  $A$  are  $-2$ ,  $1$ , and  $2$ , then the value of  $k$  is \_\_\_\_\_.

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**Solution:** The given matrix is:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix} \quad (1.2.16.1)$$

The eigenvalues of  $\mathbf{A}$  are given as:

$$\lambda_1 = -2, \quad \lambda_2 = 1, \quad \lambda_3 = 2 \quad (1.2.16.2)$$

From (1.1.18.4), the sum of the eigenvalues of a square matrix is equal to its trace:

$$\sum_{i=1}^3 \lambda_i = \text{tr}(\mathbf{A}) \quad (1.2.16.3)$$

Computing the trace of  $\mathbf{A}$ :

$$\text{tr}(\mathbf{A}) = 1 + k + (-1) = k \quad (1.2.16.4)$$

Computing the sum of the eigenvalues:

$$\sum_{i=1}^3 \lambda_i = (-2) + 1 + 2 = 1 \quad (1.2.16.5)$$

Equating the two expressions:

$$k = 1 \quad (1.2.16.6)$$

The solution can be verified from the following Python script:

```
codes/btq63/eigen.py
```